list + for loop

In [1]:
```
magicians = ['alice', 'david', 'carolina']
for magician in magicians:
    print(magician.title() + ", that was a great trick!")
    print("I can't wait to see your next trick, " + magician.title() + ".\n")
print("Thank you, everyone. That was a great magic show!")
```

```
Alice, that was a great trick!
I can't wait to see your next trick, Alice.

David, that was a great trick!
I can't wait to see your next trick, David.

Carolina, that was a great trick!
I can't wait to see your next trick, Carolina.

Thank you, everyone. That was a great magic show!
```

Excercise

In [5]:
```
# 4-1
pizzas=['beef','eggs','lamb','fish']
for pizza in pizzas:
    print(pizza.title() + " is my favorite pizza.")

print("\nPizza is the nice thing in the world." +
      "\nI ate pizza everyday,it makes me happy." +
      "\nI can't think one day i don't have pizza in the world.")

print("\nI really love pizza!")
```

```
Beef is my favorite pizza.
Eggs is my favorite pizza.
Lamb is my favorite pizza.
Fish is my favorite pizza.

Pizza is the nice thing in the world.
I ate pizza everyday,it makes me happy.
I can't think one day i don't have pizza in the world.

I really love pizza!
```

In [6]:
```
# 4-2
animals=['cat','dog','bird']
for animal in animals:
    print(animal.title() + " looks very cute!")

print("\nAny of these animals would make a great pet!")
```

```
Cat looks very cute!
Dog looks very cute!
Bird looks very cute!

Any of these animals would make a great pet!
```

list + range()

In [7]:
```
for value in range(1,5):
    print(value)
```

```
1
2
3
4
```

In [8]:
```
# numerical list
numbers=list(range(1,6))
print(numbers)
```

```
[1, 2, 3, 4, 5]
```

In [9]:
```
# even list(偶数)
even_list=list(range(2,11,2))
print(even_list)
```

```
[2, 4, 6, 8, 10]
```

In [10]:
```
# odd list(奇数)
odd_list=list(range(1,11,2))
print(odd_list)
```

```
[1, 3, 5, 7, 9]
```

In [11]:
```
# 3的倍数
multiple_3=list(range(3,21,3))
for multiple in multiple_3:
    print(multiple)
```

```
3
6
9
12
15
18
```

In [12]:
```
squares=[]
for value in range(1,11):
    square=value**2    # 注意这里是单个square没有s
    squares.append(square)

print(squares)
```

```
[1, 4, 9, 16, 25, 36, 49, 64, 81, 100]
```

List + Statistics

In [19]:
```
digits = [1, 2, 3, 4, 5, 6, 7, 8, 9, 0]
print(min(digits))
print(max(digits))
print(sum(digits))
print(len(digits))
```

```
print(sorted(digits)) # 顺序排列
print(list(reversed(digits))) # 倒序排列
```

```
0
9
45
10
[0, 1, 2, 3, 4, 5, 6, 7, 8, 9]
[0, 9, 8, 7, 6, 5, 4, 3, 2, 1]
```

List + Comprehension(列表推导式)

In [20]:
```
squares=[value**2 for value in range(1,11)]
print(squares)
```

```
[1, 4, 9, 16, 25, 36, 49, 64, 81, 100]
```

Excercise

In [38]:
```
# 4-3 for + range()
for value in range(1,6):
    print(value)
```

```
1
2
3
4
5
```

In [24]:
```
# 4-6 奇数odd
odd_numbers = list(range(1, 11, 2))
for num in odd_numbers:
    print(num)
```

```
1
3
5
7
9
```

In [21]:
```
# 4-7 multiple 3的倍数
threes = list(range(3, 21, 3))
for value in threes:
    print(value)
```

```
3
6
9
12
15
18
```

In [23]:
```
# 4-8 cube 立方
cubes=[]
for value in range(1,6):
    cubes.append(value**3)
```

```
for cube in cubes:
    print(cube)
```

```
1
8
27
64
125
```

In [25]:
```
# 4-9 Comprehension(推导式)
cubes=[value**3 for value in range(1,11)]
print(cubes)
```

```
[1, 8, 27, 64, 125, 216, 343, 512, 729, 1000]
```

list + slicing

In [27]:
```
# list 从0开始
players = ['charles', 'martina', 'michael', 'florence', 'eli']
print(players[1:3]) # 1-2 共2个(0,1)
print(players[:3]) # 0-2 共3个(0,1,2)
print(players[2:]) # 2-结尾 共3个(2,3,4)
print(players[-2:]) # -2-结尾 共2个(-2,-1)
```

```
['martina', 'michael']
['charles', 'martina', 'michael']
['michael', 'florence', 'eli']
['florence', 'eli']
```

list + slicing + for loop

In [76]:
```
players = ['charles', 'martina', 'michael', 'florence', 'eli']
print("Here are the first three players on my team:")
for player in players[:3]:
    print(player.title())
```

```
Here are the first three players on my team:
Charles
Martina
Michael
```

Copying a List

In [29]:
```
my_foods = ['pizza', 'falafel', 'carrot cake']
friend_food=my_foods[:] # 必须加冒号:

my_foods.append('ice cream')
friend_food.append('beef')
friend_food.remove('pizza')

print("My favorite foods:")
print(my_foods)

print("\nMy friend's favorite foods are:")
print(friend_food)
```

```
My favorite foods:
['pizza', 'falafel', 'carrot cake', 'ice cream']

My friend's favorite foods are:
['falafel', 'carrot cake', 'beef']
```

Excercise

In [31]:
```python
# 4-10 list + slice
players = ['charles', 'martina', 'michael', 'florence', 'eli']

# 前三个
print("The first three items in the list are:")
print(players[:3])

# 中间三个
print("\nThree items from the middle of the list are:")
print(players[1:4])

# 最后三个
print("\nThe last three items in the list are:")
print(players[-3:])
```

```
The first three items in the list are:
['charles', 'martina', 'michael']

Three items from the middle of the list are:
['martina', 'michael', 'florence']

The last three items in the list are:
['michael', 'florence', 'eli']
```

In [33]:
```python
# 4-11 list + copy
my_pizzas=['beef','eggs','lamb','fish']
friend_pizzas=my_pizzas[:]

my_pizzas.append('mango')
friend_pizzas.append('orange')

print("\nMy favorite pizzas are:")
for my_pizza in my_pizzas:
    print(my_pizza.title())

print("\nMy friend’s favorite pizzas are:")
for friend_pizza in friend_pizzas:
    print(friend_pizza.title())
```

```
My favorite pizzas are:
Beef
Eggs
Lamb
Fish
Mango

My friend’s favorite pizzas are:
Beef
Eggs
Lamb
Fish
Orange
```

In [112…

```python
# 4-12 多个list
my_pizzas=['beef','eggs','lamb','fish']
my_foods = ['pizza', 'falafel', 'carrot cake']

print("\nFirst list:")
for my_pizza in my_pizzas:
    print(my_pizza.title() + "taste very delicious.")

print("\nAnother list:")
for my_food in my_foods:
    print("I like eating " + my_food.title())
```

```
First list:
Beeftaste very delicious.
Eggstaste very delicious.
Lambtaste very delicious.
Fishtaste very delicious.

Another list:
I like eating Pizza
I like eating Falafel
I like eating Carrot Cake
```

Tuple(元组)--圆括号/不能改值、只能重新定义

In [34]:

```python
dimensions = (200, 50)
print(dimensions[0])
print(dimensions[1])
```

```
200
50
```

In [119…

```python
# 不能变更，只能重新定义
#原始tuple
dimensions1 = (200, 50)
print(dimensions1)

# 重新定义
dimensions2 = (450, 50)
print(dimensions2)
```

```
(200, 50)
(450, 50)
```

tuple + for loop

In [121…

```
dimensions = (200, 50)
for dimension in dimensions:
    print(dimension)
```

```
200
50
```

In [35]:

```
# 4-13 tuple 不能改变值
print("Original food items:")
foods=('fish','eggs','beef','lamb','chicken')
for food in foods:
    print(food.title())

# foods[0]='orange'

print("\nNew foods items:")
new_foods=('fish','eggs','tofu','lamb','shrimp')
for new_food in new_foods:
    print(new_food.title())
```

```
Original food items:
Fish
Eggs
Beef
Lamb
Chicken

New foods items:
Fish
Eggs
Tofu
Lamb
Shrimp
```

In [ ]:

In [ ]:

In [ ]:

In [ ]:

In [ ]:

In [ ]:

In [ ]: